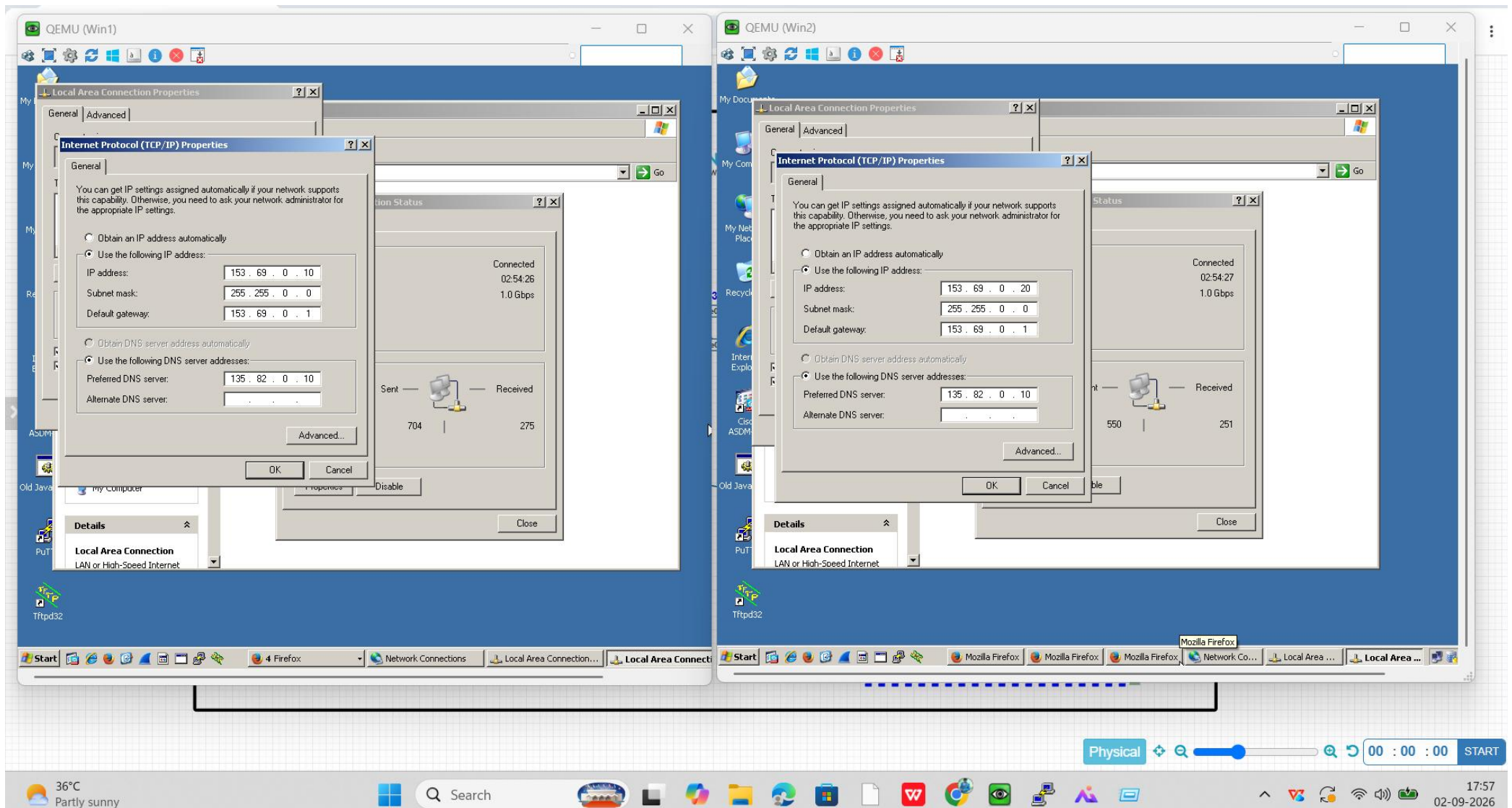


SERVER & SERVICES -PROJECT

ASSIGNED A IP ADDRESS FOR CLIENT PC'S:-



ASSIGNED A IP ADDRESS FOR MUMBAI AND CHENNAI DC'S LINUX SERVERS:-

The image displays six QEMU terminal windows, each showing the output of the `ifconfig` command for a different Linux instance. Each window shows the configuration for the `eth0` and `lo` interfaces. The `eth0` interface is configured with a unique IP address and MAC address, while the `lo` interface is configured with the standard loopback address `127.0.0.1` and MAC address `00:00:00:00:00:00`.

```
QEMU (Linux1)
# ifconfig
eth0      Link encap:Ethernet  HWaddr 00:50:00:00:01:00
          inet addr:10.0.0.10  Bcast:10.255.255.255  Mask:255.0.0.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:2456 errors:42 dropped:0 overruns:0 frame:42
          TX packets:1278 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:767998 (749.9 KiB)  TX bytes:393140 (383.9 KiB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

QEMU (Linux2)
# ifconfig
eth0      Link encap:Ethernet  HWaddr 00:50:00:00:02:00
          inet addr:10.0.0.20  Bcast:10.255.255.255  Mask:255.0.0.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:2390 errors:42 dropped:0 overruns:0 frame:42
          TX packets:1231 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:750024 (732.4 KiB)  TX bytes:381642 (372.6 KiB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

QEMU (Linux3)
# ifconfig
eth0      Link encap:Ethernet  HWaddr 00:50:00:00:06:00
          inet addr:10.0.0.30  Bcast:10.255.255.255  Mask:255.0.0.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:179 errors:0 dropped:0 overruns:0 frame:0
          TX packets:96 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:57638 (56.2 KiB)  TX bytes:30912 (30.1 KiB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

QEMU (Linux4)
# ifconfig
eth0      Link encap:Ethernet  HWaddr 00:50:00:00:03:00
          inet addr:200.10.156.10  Bcast:200.10.156.255  Mask:255.255.255.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:35 errors:0 dropped:0 overruns:0 frame:0
          TX packets:23 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:11270 (11.0 KiB)  TX bytes:7406 (7.2 KiB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

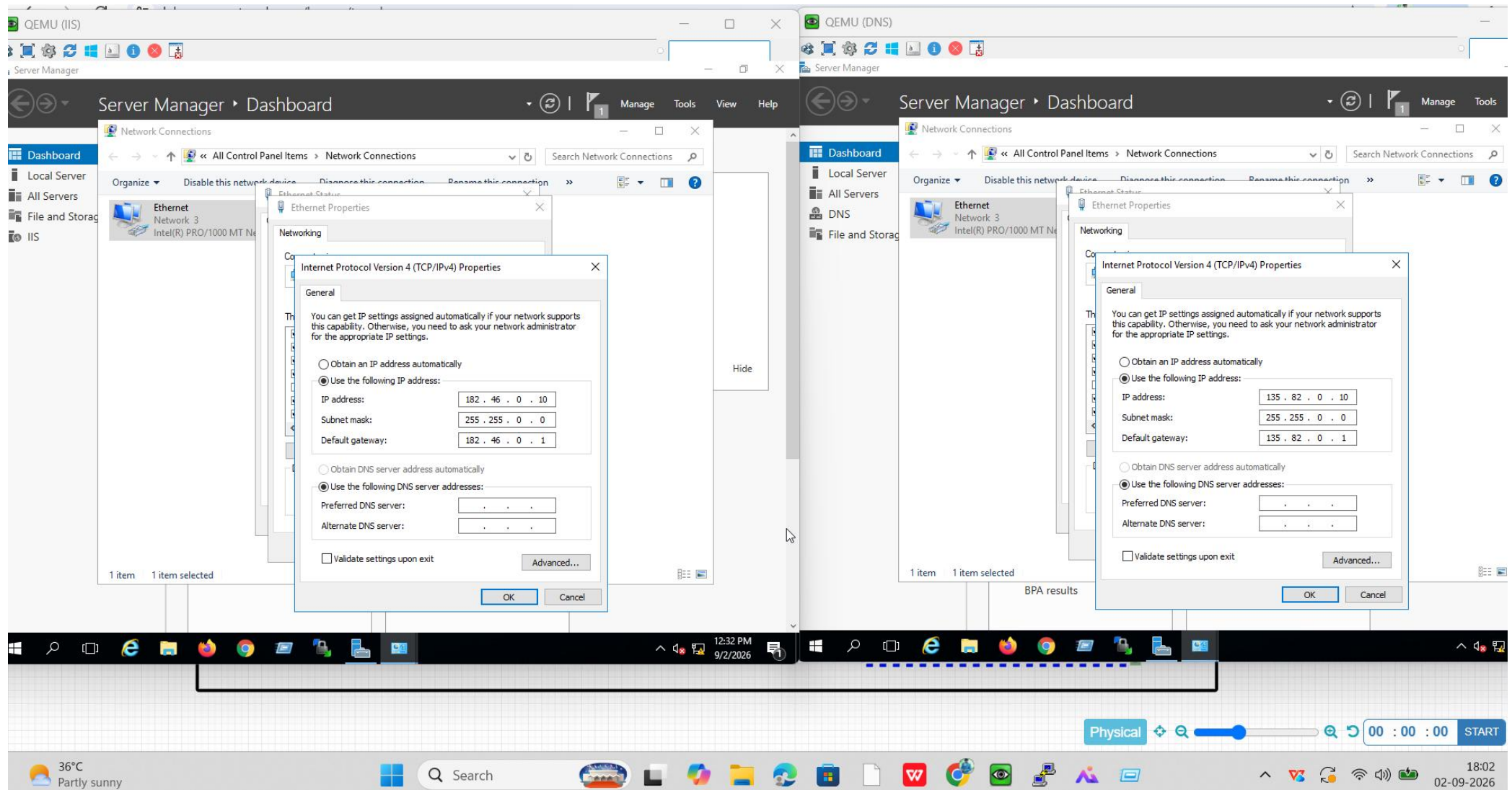
QEMU (Linux5)
# ifconfig
eth0      Link encap:Ethernet  HWaddr 00:50:00:00:04:00
          inet addr:200.10.156.20  Bcast:200.10.156.255  Mask:255.255.255.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:2138 errors:39 dropped:0 overruns:0 frame:39
          TX packets:1105 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:671054 (655.3 KiB)  TX bytes:342937 (334.0 KiB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

(Linux6)
# ifconfig
eth0      Link encap:Ethernet  HWaddr 00:50:00:00:05:00
          inet addr:200.10.156.30  Bcast:200.10.156.255  Mask:255.255.255.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:217 errors:0 dropped:0 overruns:0 frame:0
          TX packets:110 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:69874 (68.2 KiB)  TX bytes:35420 (34.5 KiB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

ASSIGNED A IP FOR PUNE DC WINDOWS SERVER AND DNS SERVER:-



ASSIGNED A IP ADDRESS FOR ALL THE CISCO ROUTERS:-

The image displays four Cisco Packet Tracer router configuration windows, each showing the output of the command `R#sh ip int br`. The windows are labeled R1, R2, R3, and R4. Each window shows a table of interface configurations.

R1 Configuration:

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	10.0.0.1	YES	manual	up	up
Ethernet0/1	1.0.0.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	up	up
Ethernet0/3	unassigned	YES	TFTP	up	up

R2 Configuration:

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	200.10.156.1	YES	manual	up	up
Ethernet0/1	2.0.0.1	YES	manual	up	up
Ethernet0/2	1.0.0.2	YES	manual	up	up
Ethernet0/3	6.0.0.1	YES	manual	up	up

R3 Configuration:

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	2.0.0.2	YES	manual	up	up
Ethernet0/1	3.0.0.1	YES	manual	up	up
Ethernet0/2	4.0.0.1	YES	manual	up	up
Ethernet0/3	unassigned	YES	TFTP	up	up

R4 Configuration:

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	182.46.0.1	YES	manual	up	up
Ethernet0/1	3.0.0.2	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	up	up
Ethernet0/3	unassigned	YES	TFTP	up	up

The bottom of the image shows a Windows taskbar with the date 02-09-2026 and time 17:37.

R5

←

R5#sh ip int br

Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	5.0.0.1	YES	manual	up	up
Ethernet0/1	4.0.0.2	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	up	up
Ethernet0/3	unassigned	YES	TFTP	up	up

R5#

R6

R6#sh ip int br

Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	135.82.0.1	YES	manual	up	up
Ethernet0/1	5.0.0.2	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	up	up
Ethernet0/3	unassigned	YES	TFTP	up	up

R6#

R7

R7#sh ip int br

Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	153.69.0.1	YES	manual	up	up
Ethernet0/1	6.0.0.2	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	up	up
Ethernet0/3	unassigned	YES	TFTP	up	up

R7#

Global-DNS

Win1

3.69.0.10

Win2

53.69.0.20

Physical

00 : 00 : 00

START

36°C Partly sunny

Search

17:41 02-09-2026

MANUALLY ADD A STATIC ROUTE FOR ALL THE MISSING ROUTES AT CISCO ROUTER'S (R1-R7)

The image displays four Cisco IOS command-line windows (R1, R2, R3, R4) showing network configuration and routing tables. The windows are arranged in a 2x2 grid. The bottom of the image shows a Windows taskbar with the date 02-09-2026 and time 17:45.

R1:

```
+ - replicated route, % - next hop override, p - overrides from PfR
Gateway of last resort is not set

  1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       1.0.0.0/8 is directly connected, Ethernet0/1
L       1.0.0.1/32 is directly connected, Ethernet0/1
S       2.0.0.0/8 [1/0] via 1.0.0.2
S       3.0.0.0/8 [1/0] via 1.0.0.2
S       4.0.0.0/8 [1/0] via 1.0.0.2
S       5.0.0.0/8 [1/0] via 1.0.0.2
S       6.0.0.0/8 [1/0] via 1.0.0.2
  10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Ethernet0/0
L       10.0.0.1/32 is directly connected, Ethernet0/0
S      135.82.0.0/16 [1/0] via 1.0.0.2
S      153.69.0.0/16 [1/0] via 1.0.0.2
S      182.46.0.0/16 [1/0] via 1.0.0.2
S      200.10.156.0/24 [1/0] via 1.0.0.2
R1#
R1#
R1#
R1#
R1#
```

R2:

```
Gateway of last resort is not set

  1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
O/C     1.0.0.0/8 is directly connected, Ethernet0/2
E/L     1.0.0.2/32 is directly connected, Ethernet0/2
  2.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
EC      2.0.0.0/8 is directly connected, Ethernet0/1
L       2.0.0.1/32 is directly connected, Ethernet0/1
E       3.0.0.0/24 is subnetted, 1 subnets
S       3.0.0.0 [1/0] via 2.0.0.2
  4.0.0.0/24 is subnetted, 1 subnets
S       4.0.0.0 [1/0] via 2.0.0.2
  5.0.0.0/24 is subnetted, 1 subnets
S       5.0.0.0 [1/0] via 2.0.0.2
  6.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       6.0.0.0/8 is directly connected, Ethernet0/3
L       6.0.0.1/32 is directly connected, Ethernet0/3
S      10.0.0.0/8 [1/0] via 1.0.0.1
S      135.82.0.0/16 [1/0] via 2.0.0.2
S      153.69.0.0/16 [1/0] via 6.0.0.2
S      182.46.0.0/16 [1/0] via 2.0.0.2
  200.10.156.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.10.156.0/24 is directly connected, Ethernet0/0
L       200.10.156.1/32 is directly connected, Ethernet0/0
```

R3:

```
Gateway of last resort is not set

  1.0.0.0/24 is subnetted, 1 subnets
S       1.0.0.0 [1/0] via 2.0.0.1
  2.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       2.0.0.0/8 is directly connected, Ethernet0/0
L       2.0.0.2/32 is directly connected, Ethernet0/0
  3.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       3.0.0.0/8 is directly connected, Ethernet0/1
L       3.0.0.1/32 is directly connected, Ethernet0/1
  4.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       4.0.0.0/8 is directly connected, Ethernet0/2
L       4.0.0.1/32 is directly connected, Ethernet0/2
  5.0.0.0/24 is subnetted, 1 subnets
S       5.0.0.0 [1/0] via 4.0.0.2
  6.0.0.0/24 is subnetted, 1 subnets
S       6.0.0.0 [1/0] via 2.0.0.1
S      10.0.0.0/8 [1/0] via 2.0.0.1
S      135.82.0.0/16 [1/0] via 4.0.0.2
S      153.69.0.0/16 [1/0] via 2.0.0.1
S      182.46.0.0/16 [1/0] via 3.0.0.2
S      200.10.156.0/24 [1/0] via 2.0.0.1
R3#
```

R4:

```
Gateway of last resort is not set

  1.0.0.0/24 is subnetted, 1 subnets
S       1.0.0.0 [1/0] via 3.0.0.1
  2.0.0.0/24 is subnetted, 1 subnets
S       2.0.0.0 [1/0] via 3.0.0.1
  3.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       3.0.0.0/8 is directly connected, Ethernet0/1
L       3.0.0.2/32 is directly connected, Ethernet0/1
  4.0.0.0/24 is subnetted, 1 subnets
S       4.0.0.0 [1/0] via 3.0.0.1
  5.0.0.0/24 is subnetted, 1 subnets
S       5.0.0.0 [1/0] via 3.0.0.1
  6.0.0.0/24 is subnetted, 1 subnets
S       6.0.0.0 [1/0] via 3.0.0.1
S      10.0.0.0/8 [1/0] via 3.0.0.1
S      135.82.0.0/16 [1/0] via 3.0.0.1
S      153.69.0.0/16 [1/0] via 3.0.0.1
  182.46.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       182.46.0.0/16 is directly connected, Ethernet0/0
L       182.46.0.1/32 is directly connected, Ethernet0/0
S      200.10.156.0/24 [1/0] via 3.0.0.1
R4#
```



```

S      1.0.0.0/24 is subnetted, 1 subnets
        1.0.0.0 [1/0] via 4.0.0.1
S      2.0.0.0/24 is subnetted, 1 subnets
        2.0.0.0 [1/0] via 4.0.0.1
S      3.0.0.0/24 is subnetted, 1 subnets
        3.0.0.0 [1/0] via 4.0.0.1
C      4.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L      4.0.0.0/8 is directly connected, Ethernet0/1
L      4.0.0.2/32 is directly connected, Ethernet0/1
C      5.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      5.0.0.0/8 is directly connected, Ethernet0/0
L      5.0.0.1/32 is directly connected, Ethernet0/0
S      6.0.0.0/24 is subnetted, 1 subnets
        6.0.0.0 [1/0] via 4.0.0.1
S      10.0.0.0/8 [1/0] via 4.0.0.1
S      135.82.0.0/16 [1/0] via 5.0.0.2
S      153.69.0.0/16 [1/0] via 4.0.0.1
S      182.46.0.0/16 [1/0] via 4.0.0.1
S      200.10.156.0/24 [1/0] via 4.0.0.1

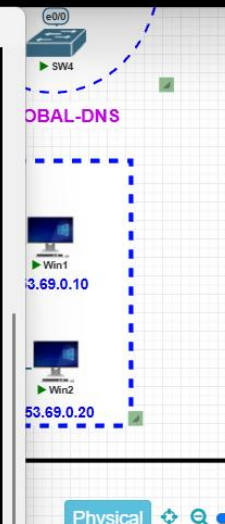
```

```
S 1.0.0.0/24 is subnetted, 1 subnets
    1.0.0.0 [1/0] via 5.0.0.1
S 2.0.0.0/24 is subnetted, 1 subnets
    2.0.0.0 [1/0] via 5.0.0.1
S 3.0.0.0/24 is subnetted, 1 subnets
    3.0.0.0 [1/0] via 5.0.0.1
S 4.0.0.0/24 is subnetted, 1 subnets
    4.0.0.0 [1/0] via 5.0.0.1
S 5.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 5.0.0.0/8 is directly connected, Ethernet0/1
L 5.0.0.2/32 is directly connected, Ethernet0/1
S 6.0.0.0/24 is subnetted, 1 subnets
    6.0.0.0 [1/0] via 5.0.0.1
S 10.0.0.0/8 [1/0] via 5.0.0.1
S 135.82.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 135.82.0.0/16 is directly connected, Ethernet0/0
L 135.82.0.1/32 is directly connected, Ethernet0/0
S 153.69.0.0/16 [1/0] via 5.0.0.1
S 182.46.0.0/16 [1/0] via 5.0.0.1
S 200.10.156.0/24 [1/0] via 5.0.0.1
```

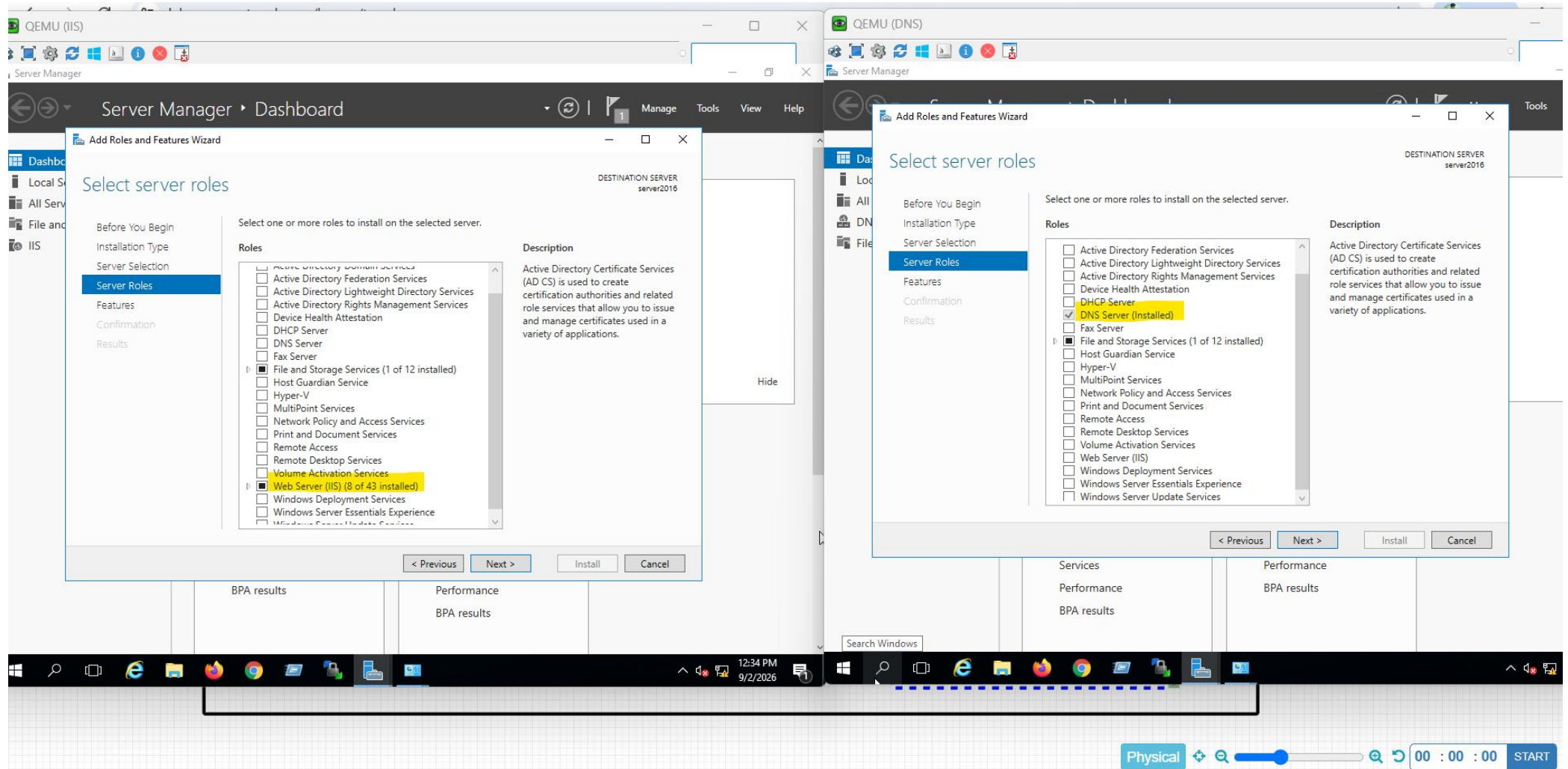
```

S      1.0.0.0/24 is subnetted, 1 subnets
      1.0.0.0 [1/0] via 6.0.0.1
S      2.0.0.0/24 is subnetted, 1 subnets
      2.0.0.0 [1/0] via 6.0.0.1
S      3.0.0.0/24 is subnetted, 1 subnets
      3.0.0.0 [1/0] via 6.0.0.1
S      4.0.0.0/24 is subnetted, 1 subnets
      4.0.0.0 [1/0] via 6.0.0.1
S      5.0.0.0/24 is subnetted, 1 subnets
      5.0.0.0 [1/0] via 6.0.0.1
C      6.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L      6.0.0.0/8 is directly connected, Ethernet0/1
S      6.0.0.2/32 is directly connected, Ethernet0/1
S      10.0.0.0/8 [1/0] via 6.0.0.1
S      135.82.0.0/16 [1/0] via 6.0.0.1
C      153.69.0.0/16 is variably subnetted, 2 subnets, 2 masks
L      153.69.0.0/16 is directly connected, Ethernet0/0
S      153.69.0.1/32 is directly connected, Ethernet0/0
S      182.46.0.0/16 [1/0] via 6.0.0.1
S      200.10.156.0/24 [1/0] via 6.0.0.1

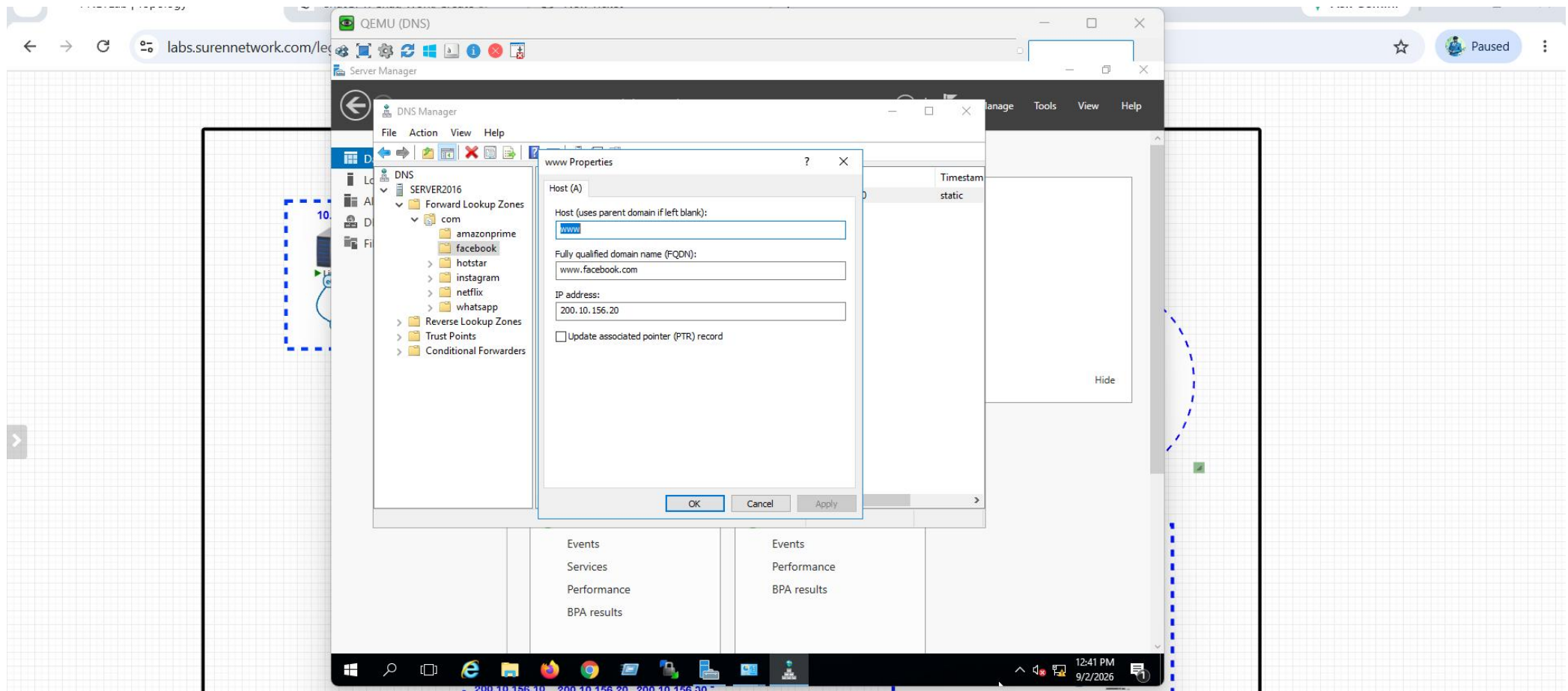
```



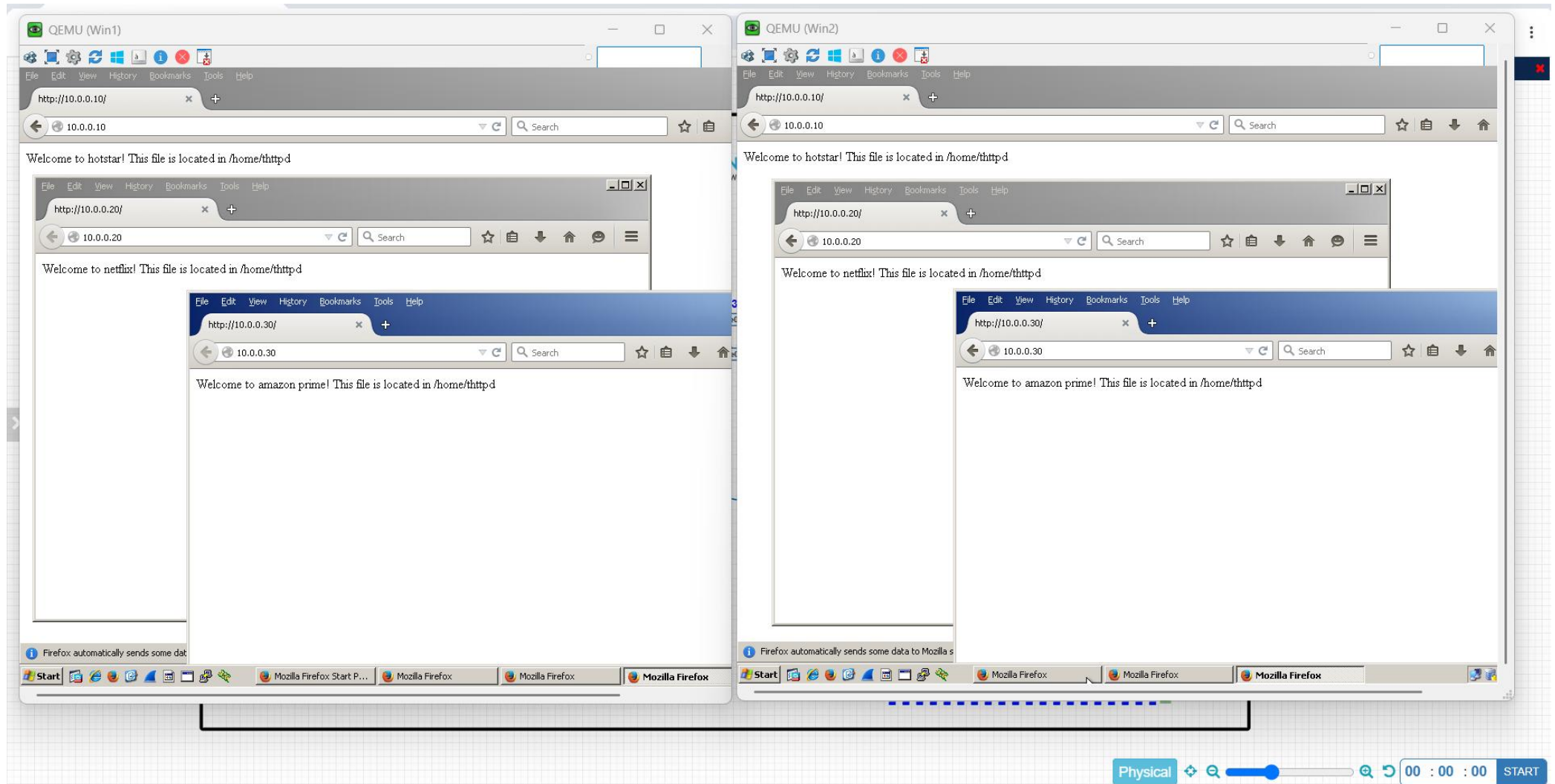
INSTALL A WEBSERVICE IISS AT PUNE DC AND INSTALL DNS SERVICE AT GLOBAL DNS SERVER:-



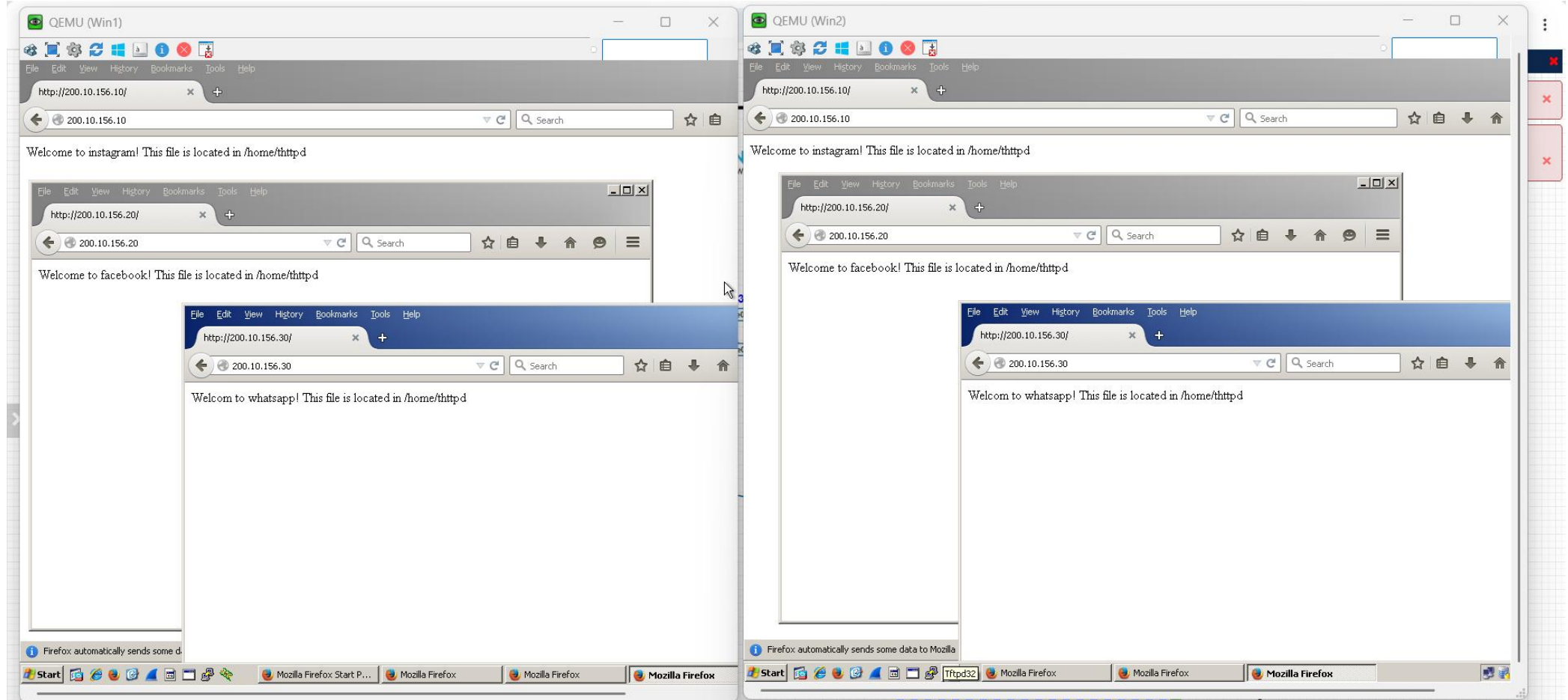
AFTER INSTALLED A DNS SERVICE CREATE A DNS RECORD FOR ALL THE DC'S LINUX AND WINDOWS SERVERS:-



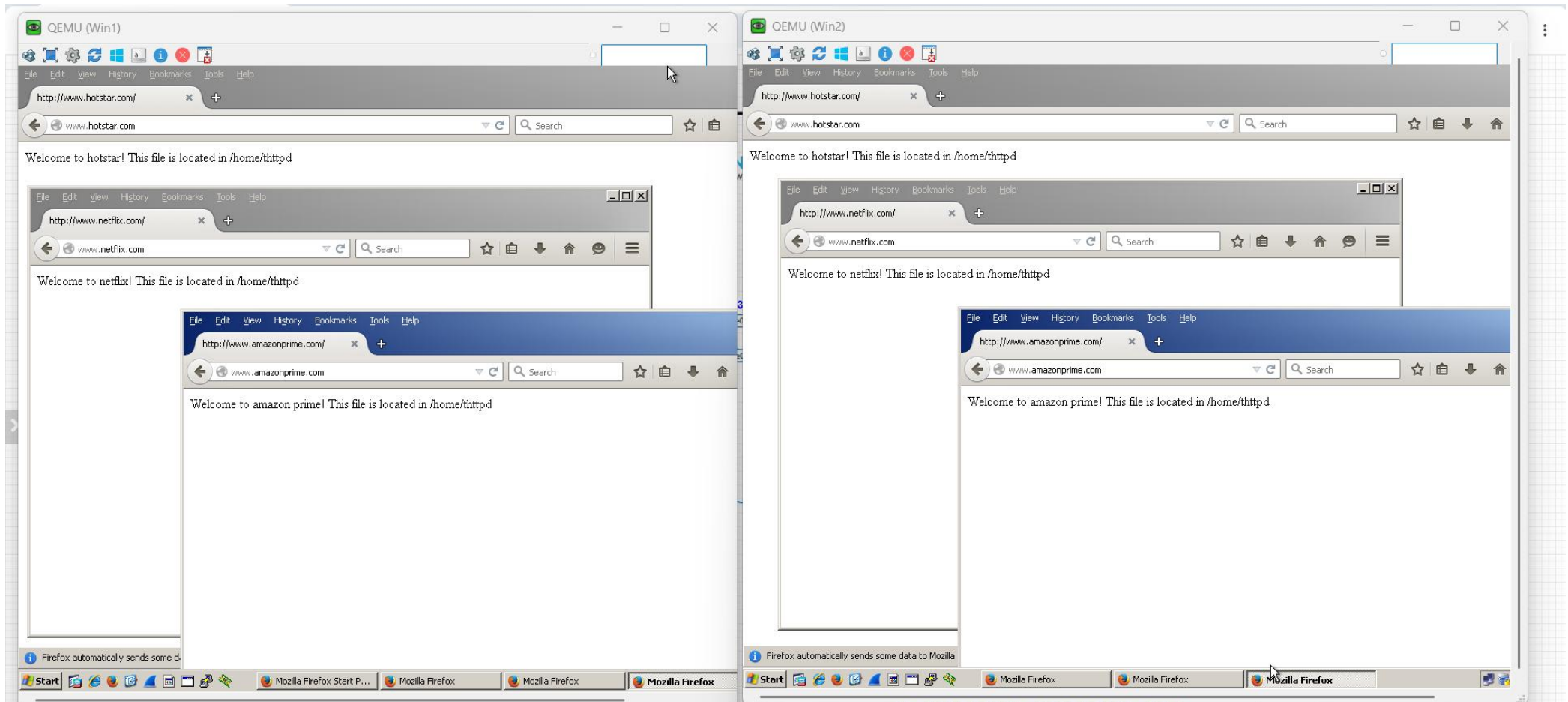
BOTH CLIENT PC'S TO MUMBAI DC SERVER'S REACHED VIA IP ADDRESS OUTPUT:-



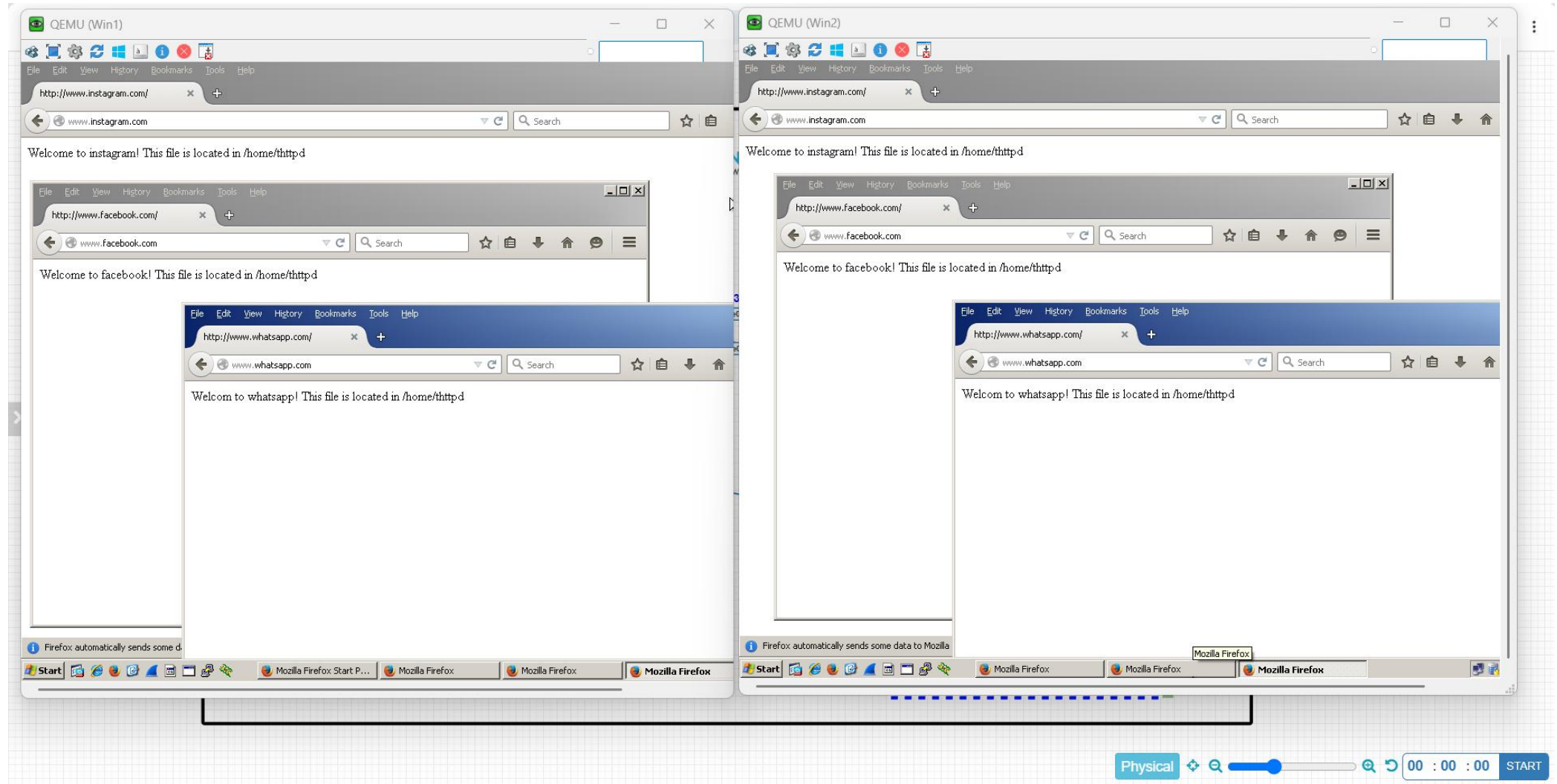
BOTH CLIENT PC'S TO CHENNAI DC SERVER'S REACHED VIA IP ADDRESS OUTPUT:-



BOTH CLIENT PC'S TO MUMBAI DC SERVER'S REACHED VIA DOMAIN NAME OUTPUT:-



BOTH CLIENT PC'S TO CHENNAI DC SERVER'S REACHED VIA DOMAIN NAME OUTPUT:-



BOTH CLIENT PC'S TO PUNE DC SERVER'S REACHED VIA IP ADDRESS & DOMAIN NAME OUTPUT:-

